

AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
305 A	17.49 MVA	90.0 degC	81.0 degC	20.186 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Al	
Conductor size		185	mm2
Conductor type		stranded	
Conductor diameter Dc		15.9	mm
Conductor screen OD Dc_sc		17.6	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		35.9	mm
Ins outer screen OD Di_sc		37.6	mm
Insulation thickness t_ins		9.19	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
Screen type		Cu_tape	
Screen mean diameter ds		38.0	mm
Screen area As		25.0	mm2
Armour type		none	
Oversheath material		PVC	
Cable outer diameter De		44.0	mm
Rated voltage U		33.0	kV
U0 line-to-earth		19.000	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1200	mm
Axial spacing s		44.0	mm
Ambient ground temp theta_a		30	degC
Soil thermal resistivity rho_soil		1.20	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		1.6400e-04	ohm/m
Temperature coefficient a20		4.0300e-03	1/K
R0 at theta_amb = 30 degC		1.7061e-04	ohm/m
R0 at theta_max = 90 degC		2.1026e-04	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
$xs4 = [ks(8\pi i f / R0) \times 10^{-7}]^2$		0.357181	
ys skin effect factor		0.001858	
$xp4 = [kp(8\pi i f / R0) \times 10^{-7}]^2$		0.228596	
yp" intermediate proximity		0.001189	
yp proximity factor		0.000682	
R at theta_max AC = $R0(1+ys+yp)$		2.1080e-04	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
U0 line-to-earth		19.0000	kV
Capacitance C = $er / (18 \ln(Di_sc / Dc_sc)) \times 10^{-9}$		0.182964	nF/m
$\omega = 2 \pi f$		314.1593	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		8.3001e-02	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		25.00	mm2
Lay factor LFs		1.000000	
Screen Rs0 at 20 degC (with LFs)		6.8964e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_sc semiconducting screen		2.50	K.m/W
rho_tape bedding tape		6.00	K.m/W
T1_cs conductor screen		0.040417	K.m/W
T1_i insulation		0.397081	K.m/W
T1_si insulation screen		0.018409	K.m/W
T1_uwbt bedding tape under wire screen		0.000000	K.m/W
T1_raw = sum of layers		0.455907	K.m/W
T1 correction factor x1.00 (wire coverage check)		0.455907	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
T3_raw oversheath		0.123865	K.m/W
T3 x1.6 touching trefoil factor		0.198184	K.m/W
u = 2L/De		54.5455	
T4 soil		2.327459	K.m/W
Sum T = T1+T2+T3+T4		2.981550	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		80.9838	degC
Rs at theta_s		8.5492e-04	ohm/m
X sheath reactance		5.2763e-05	ohm/m
lam1' circulating current		0.015389	
lam1" eddy current		0.003291	
lam1 = lam1' + lam1"		0.018680	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		60.00	K
Etop numerator		59.771449	
Ebot denominator		6.3845e-04	
I = sqrt(Etop/Ebot) exact		305.9729	A
I rounded TB880 GP2		305	A
MVA = sqrt(3) x U x I		17.489	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		80.9838	degC
theta_j cable outer surface		76.9832	degC
theta_a ambient ground		30.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I ² x R		19.734827	W/m
Sheath Ws = lam1 x Wc		0.368650	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		8.3001e-02	W/m
Total		20.186477	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.167950	uH/m
Reactance X = omega L		52.7631	uOhm/m
Pos-seq resistance R1		214.7362	uOhm/m
Pos-seq impedance Z1		221.1234	uOhm/m
Capacitance C		0.182964	nF/m
Charging current Ic		1.0921	mA/m

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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